

CAISO Data Spec — Target Plant Bid-Gap Analysis

Goal

For any target plant, determine what it actually bid into the market, what it would have needed to bid to clear, and how often/by how much it missed. This is the input the subsidy-sizing model needs — how the data gets structured and pulled is an implementation choice.

What we need, conceptually

- **Identity:** every CAISO dataset is keyed by Resource ID / PNode, not plant name. Any plant we want to analyze needs to be resolved to its Resource ID first, via CAISO's Master File.
- **Bids:** what the plant actually submitted, hour by hour — this is the resource's real cost/willingness curve. This is confidential in near-real-time and becomes public after a 90-day delay (CAISO's Public Bid Data). Historical analysis doesn't need faster access than that.

- **Clearing outcomes:** what the market price actually was, and whether the plant's bid was low enough to clear. This is public with no delay.
- **Comparison context:** a bid only matters relative to other resources competing to relieve the same local constraint — a statewide average price isn't the right benchmark. Whatever comparison set gets built should be scoped to the plant's local capacity/transmission area, not type-matched nationally.
- **Outage awareness:** a plant not clearing because it was offline is a different fact than a plant not clearing because it was priced out. Distinguishing these matters for interpreting any "missed" hour.

What determines whether a subsidy would work, once the data exists

- Hours where the plant's bid came close to the clearing price but didn't quite make it are the meaningful target — a subsidy sized to close a small gap is cheap and effective.
- Hours with a large gap between bid and clearing price aren't worth modeling for subsidy purposes — no plausible subsidy closes them.

- Hours where the plant already cleared, or was self-scheduled (committed to run regardless of price), aren't subsidy targets — there's nothing to influence there.

Access considerations

- Public Bid Data and clearing price data are both retrievable through CAISO's OASIS system. Some CAISO datasets (LMP, load, curtailment, and others) are already wrapped by existing open-source tooling (e.g., the `gridstatus` Python package) that handles the underlying query format — worth checking what's already covered before writing raw API calls. Public Bid Data specifically may not be one of the pre-wrapped datasets, in which case it needs a direct OASIS query — the exact report name and version should be confirmed against CAISO's current interface documentation rather than assumed, since it has changed across spec versions.
- Real-time (non-delayed) visibility into a plant's bid is only possible for the plant's own Scheduling Coordinator — no path exists to see another plant's bids sooner than the 90-day public release, and none is needed for historical analysis.

Open item

The exact Public Bid Data report identifier needs confirming against the current CAISO OASIS spec before building against it — this is the one unverified piece.

Map Interface — Candidate Plant Viewer

Goal

A way to see where candidate plants are, with their key attributes, without introducing cost or API-key dependencies.

What the interface needs to do

- Show plant locations from the existing plant database (coordinates, capacity, prime mover type, CAISO dispatch flag, distance to coast, water source), and later, whatever bid-gap or capacity-factor results come out of the analysis above.
- Let someone distinguish between plant categories at a glance (e.g., strong host candidates vs. subsidy-dependent candidates vs. rejected).
- Let someone drill into a specific plant's details.

- Filter or narrow the set by attribute (capacity, prime mover type, dispatch status) when the full list is too much to look at at once.

Constraints

- No paid mapping API or per-request cost — this should run on free map tiles and a free mapping library (options exist that require no API key at all).
- No dependency on a hosted backend if avoidable — a self-contained file that reads the plant database directly is preferable to standing up infrastructure for what's fundamentally a static reference view.
- Should be easy to regenerate or update as the plant database or analysis results change, rather than a one-off hand-built map.

Open item

Whether the map needs to reflect live/updating data (e.g., refreshed automatically as new bid-gap results come in) or is acceptable as a periodically-regenerated static view — that affects whether a lightweight backend is worth the added complexity.